

Ex. No.: 5

System Calls Programming

Aim: To experiment system calls using fork(), execlp() and pid() functions.

Algorithm:

1. Start

- o Include the required header files (stdio.h and stdlib.h).

2. Variable Declaration

- o Declare an integer variable pid to hold the process ID.

3. Create a Process

- o Call the fork() function to create a new process. Store the return value in the pid variable:

If fork() returns:

-1: Forking failed (child process not created).

0: Process is the child process.

Positive integer: Process is the parent process.

4. Print Statement Executed Twice

- o Print the statement:

scss

Copy code

THIS LINE EXECUTED TWICE

(This line is executed by both parent and child processes after fork()).

5. Check for Process Creation Failure

- o If pid == -1:

Print:

Copy code

CHILD PROCESS NOT CREATED

Exit the program using exit(0).

6. Child Process Execution

o If pid == 0 (child process):

Print:

Process ID of the child process using getpid().

Parent process ID of the child process using getppid().

7. Parent Process Execution

o If pid > 0 (parent process):

Print:

Process ID of the parent process using getpid().

Parent's parent process ID using getppid().

8. Final Print Statement

o Print the statement:

objectivec

Copy code

IT CAN BE EXECUTED TWICE

(This line is executed by both parent and child processes).

9. End

```
1 #include <stdio.h>
2 #include <unistd.h>
3 #include <dirent.h>
4 #include <stdlib.h>
5 int main()
6 {
7     int pid;
8     pid = fork();
9     printf("\nTHIS LINE EXECUTED TWICE");
10    if(pid==0)
11    {
12        printf("\nCHILD PROCESS NOT CREATED");
13        exit(0);
14    }
15    if(pid == 0)
16    {
17        printf("\nCHILD PROCESS CREATED AND ID IS: %d\n",getpid());
18        printf("\nCHILD'S PARENT PROCESS ID IS: %d\n",getppid());
19    }
20    else
21    {
22        printf("\nPARENT PROCESS CREATED AND ID IS: %d\n",getpid());
23        printf("\nPARENT'S PARENT PROCESS ID IS: %d\n",getppid());
24    }
25    printf("\nIT CAN BE EXECUTED TWICE\n");
26    return 0;
27 }
28 |
```

```
[PraveenKumar212@localhost ~]$ cd system_calls  
[PraveenKumar212@localhost system_calls]$ gcc fork.c -o fork  
[PraveenKumar212@localhost system_calls]$ ./fork
```

```
THIS LINE EXECUTED TWICE  
PARENT PROCESS CREATED AND ID IS: 34307
```

```
PARENT'S PARENT PROCESS ID IS: 34266
```

```
IT CAN BE EXECUTED TWICE
```

```
THIS LINE EXECUTED TWICE  
CHILD PROCESS CREATED AND ID IS: 34308
```

```
CHILD'S PARENT PROCESS ID IS: 2403
```

```
IT CAN BE EXECUTED TWICE  
[PraveenKumar212@localhost system_calls]$
```